

COMPARING PERFORMANCE OF MAMBA AND SELF-ATTENTION MODELS ON IMBALANCED IMAGE CLASSIFICATION TASKS

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ABSTRACT

Deep learning has made huge contributions in the medical field, producing models that can analyze medical images, predict development of diseases, and tailor treatment plans. Unfortunately, datasets used are frequently imbalanced, which can lead to biased and worse model performance. The Mamba architecture is relatively new and is a foundation of many models, but the Mamba component’s ability to adapt to imbalanced data has not been explored much. Our study is one of the first to evaluate how well Mamba can perform on imbalanced datasets and compare it to a popular architecture, Transformer. I test on various traditional and medical datasets spanning a range of imbalance ratios and find that, in general, the Mamba architecture is able to outperform the Transformer architecture. The Transformer models require a stronger data augmentation to be competitive with Mamba, suggesting Mamba’s better ability to perform under imbalance.

Index Terms— deep learning, data imbalance, medical image classification

1. INTRODUCTION

Datasets are often imbalanced, falling into a long-tailed distribution [1]. This problem is prevalent in many of medical datasets, which are crucial in training deep learning models for tasks like identifying diseases and conditions. The overall goal is to provide better patient outcomes by giving providers a better understanding of the patient; however, imbalanced datasets negatively affect a model’s performance [2, 3] which can harm the patient, leading to severe consequences. There are many different model architectures for image classification, one of the most popular being Vision Transformers [4]. But, there’s a new architecture that has been on the rise lately: Mamba [5]. The newly introduced Mamba has shown strong performances in many modalities and has started to be used for healthcare datasets [6]. Despite the increasing popularity of Mamba, there is little work that has explored on how the architecture performs on long-tailed data, especially in comparison against its competitor, the Vision Transformer (ViT). Therefore, it is important to evaluate the two architectures to determine in crucial settings, like healthcare, where imbalanced data may be very common, which models perform bet-

ter.

Given the lack of work that investigates the performance of Mamba and Vision Transformers in various levels of data imbalance, especially in the healthcare setting, we make the following contributions:

1. **Comparing ViT and Mamba models on various levels of artificially imbalanced CIFAR-10 and CIFAR-100 [7] datasets:** We evaluate the Top-1 accuracy of DEiT-TINY and ViM-TINY on balanced and imbalanced levels, i.e., $10\times$, and $100\times$, of CIFAR-10 and CIFAR-100 and find that ViM-Ti outperforms DEiT-Ti in all experiments. We then apply a stronger image augmentation regime on DEiT-Ti and discover only then is the model competitive with the baseline ViM-Ti.
2. **Comparing ViT and Mamba models on select MedMNIST2D [8] datasets:** We continue to evaluate the Top-1 accuracy of the same models on certain datasets within MedMNIST2D and find that ViM-Ti outperforms default DEiT-Ti on almost all the datasets. Like before, the augmented DEiT-Ti was able to outperform ViM-Ti in several of the datasets.

2. RELATED WORK

In machine learning, the occurrence of data imbalance, known as an imbalanced or long-tailed dataset, is a common problem as the unequal distribution of classes can lead to worse overall performance [2]. In healthcare, the impact of imbalanced datasets results in imperfections that can lead to a biased training in models and more false positives and negatives, resulting in heavy consequences, such as adversely affecting a patient’s outcome [3].

Many computer vision tasks have been dominated by self-attention-based models, like the Vision Transformer, that use self-attention to classify images after they have been split into a series of patches and processed [9, 4, 10]. Recently, structured state space models (SSMs), have been introduced as an alternative to self-attention [11]. They are a core component of Mamba, a new model architecture made to address the limitations of Transformers, and has achieved state-of-the-art performance on several data-types in addition to having sub-quadratic time complexity and faster inference [5]. Recently, Mamba has been adapted for images [12, 13, 14]. Al-

MODEL (# PARAMS)	CIFAR-10			CIFAR-100		
	IR = 1×	10×	100×	1×	10×	100×
DEiT-TINY (6M)	0.845	0.627	0.433	0.542	0.348	0.226
DEiT-TINY+ (6M)	0.878	0.685	0.483	0.593	0.408	0.262
VIM-TINY (7M)	0.914	0.754	0.543	0.595	0.370	0.254

Table 1. Top-1 accuracy of a Mamba and Transformer-based model on varying levels of imbalance ratio (IR). DEiT-TI+ denotes a model trained with a stronger augmentation regime than DEiT-TI.

though both architectures have achieved strong performances, the way the models learn differs greatly.

Both types of architectures are now being used in the medical field. Mamba-based models have been shown to capture small details and are able to extract the important features and information, which is critical for medical tasks [6]. Thus, many models have been released for the field, and many have unique changes to their architecture and combine components, like convolutional layers, to improve capturing of local features for long-ranged modelings [15, 16, 17]. Similarly, ViTs have been widely established to perform a variety of tasks such as image reconstruction, image captioning, and three-dimensional imaging [18, 19].

Given the two architectures’ widespread adoption by the medical field, it is crucial to evaluate how well both these architectures perform on imbalanced medical datasets due to the impact they may have on patients’ outcome. Our work will explore the more basic Vision Transformer and Vision Mamba instead of the medical field variants to reduce the confounding of extra architectural components.

3. METHODOLOGY

3.1. Datasets and Metrics

We first use CIFAR-10 and CIFAR-100 [7] and artificially induce data imbalance by creating a subset of the dataset via sampling. We use an exponential function

$$n_i = N * \frac{1}{\text{IR}^{\frac{i}{\# \text{ classes}}}}$$

where n_i is the number of samples for class i (i starts from zero), N is the original number of samples per class, and IR (imbalance ratio) is the ratio between the number of samples for the rarest class and that of the most common. We choose a subset of the MedMNIST2D dataset collection [8]: DermaMNIST (Derma), BloodMNIST (Blood), OrganAMNIST (OrganA), OrganCMNIST (OrganC), and OrganSMNIST (OrganS), based on their “multi-class” classification task and size of less than 100,000 total samples due to computational constraints. We report the Top-1 accuracy on the test set. For CIFAR-10 and CIFAR-100, we set aside 10% of the training data for the validation set, while the MedMNIST datasets maintains their own train, validation, and test split.

3.2. Training

We train a variant of the Vision Transformer [4], the Data Efficient Image Transformer (DeiT) [20] and its Mamba-architecture-based alternative Vision Mamba (Vim) [12] at the $< 10M$ parameter scale: DEiT-TINY and VIM-TINY. We choose Vision Mamba due to its architectural similarities with Vision Transformers. All models were trained from scratch for 300 epochs with a batch size of 128, using a cosine scheduler with a linear 15 epoch warmup (5% of the entire training) and regular cross entropy loss. We also use an AdamW optimizer with $\beta = (0.9, 0.999)$ and a 0.01 weight decay. For the base CIFAR and MedMNIST datasets, we search through $\text{lr} \in \{1\text{e-}4, 2.5\text{e-}4, 5\text{e-}4, 1\text{e-}3, 2.5\text{e-}3, 5\text{e-}3\}$ to determine the best learning rate based on the validation accuracy. For the imbalanced versions of CIFAR-10 and 100 ($\text{IR} = 10\times, 100\times$), we train using the learning rate determined from the base dataset. All images were resized to 224 pixels to work with the models.

For DEiT-TI and VIM-TI, we use very little image augmentation compared to traditional training procedures, electing to use random resized crop and random horizontal flip along with normalization. For CIFAR-10 and 100, we use the dataset’s mean and standard deviation, and for the colored MedMNIST datasets, we use ImageNet normalization values [21] and use (0.5, 0.5) for the gray-scale datasets [8]. We also applied stronger image augmentation onto DeiT, resulting in DEiT-TI+. The procedure was based on the original DeiT training process [20], utilizing color jitter (0.3), auto augmentation (*rand-m9-mstd0.5-inc1*), and random pixel erase (0.25), while excluding Mixup and Cutup.

4. EXPERIMENTS

4.1. CIFAR

We begin by analyzing the standard ability of the two different classes of architectures on standard CIFAR-10 and CIFAR-100 as the baseline. We noticed the large gap between DEiT-TI and VIM-TI which prompted us to introduce DEiT-TI+ which had a stronger training procedure 3.2, which closed the gap between the accuracies. The DEiT-TI+ results are still a bit lower than other reported values for base CIFAR-10 and 100, which might be due to the need of a strong data augmen-

DATASET	# Classes	TRAIN SPLIT IR	VALIDATION SPLIT IR	TEST SPLIT IR
BLOOD	8	2.74	2.73	2.74
ORGANA	11	4.54	4.59	4.19
ORGANC	11	5.02	4.47	4.36
ORGANS	11	5.64	5.17	5.23
DERMA	7	58.66	55.92	58.30

Table 2. Imbalance ratio (IR) of selected multi-class classification-based datasets with under 100,000 samples within MedMNIST2D. These datasets are used later to evaluate the performance of Mamba and Transformer architectures.

MODEL (# PARAMS)	BLOOD	ORGANA	ORGANC	ORGANS	DERMA
DEiT-TINY (6M)	0.968	0.873	0.871	0.734	0.753
DEiT-TINY+ (6M)	0.963	0.907	0.896	0.758	0.782
VIM-TINY (7M)	0.968	0.910	0.880	0.764	0.780

Table 3. Top-1 accuracy of Mamba and Transformer-based model on select multi-class classification MedMNIST2D datasets. DEiT-TI+ denotes a model trained with a stronger augmentation regime.

tation regime for Vision Transformers [22]. However, to the best of our knowledge, our work has the highest Vim accuracies on CIFAR-10/100. We report the results in Table 1.

4.2. MedMNIST

Given our results on our base datasets, we test the two architecture types of imbalanced, healthcare-related data. We use datasets from MedMNIST because of its easy accessibility and dataset variety. We select a mix of colored and gray-scaled datasets that were multi-class classifications tasks because of its similarity to CIFAR and it satisfied requirements detailed in Section 3.2. The imbalance ratios for the chosen datasets are in Table 2. For all dataset and model combinations, all learning rates mentioned in 3.2 were tested with the best validation model contributing to the Top-1 testing accuracy values in Table 3.

5. DISCUSSION AND CONCLUSIONS

We discover in the experiments that VIM-TINY outperform DEiT-TINY on all imbalance ratios of both CIFAR datasets. During evaluations, we also noticed that VIM-TI overfit less than DEiT-TI, which could be due to Mamba’s inability to directly attend to the inputs compared to self-attention’s ability. Only after adding stronger image augmentations to DEiT-TI was it able to outperform its Mamba-based counterpart on select imbalance ratios while seeing an overall improvement in performance across the board.

In actual medical datasets, VIM-TI outperforms DEiT-TI on all datasets, except for Blood where they were equivalent. Our experiments highlight the importance of augmentation for Transformer-based models, as after augmentation, DEiT-

TI+ and VIM-TI were now competitive. For DEiT-TI+ and VIM-TI, the average gap between the colored datasets was 0.35 percentage points compared to 0.83 for the gray-scaled datasets. The higher variance on gray-scaled datasets shows the need for future users to run thorough experiments when comparing these two architectures on similar datasets, while for colored datasets, they perform quite similarly. We highlight that, compared to [8]’s benchmarks that use the CNN-based ResNet, self-attention and Mamba-based models performed better on the colored datasets while the CNN-based models performed better on gray-scaled datasets. However, there are differences in training procedure between our work and theirs, namely, training epochs, optimizer, augmentation, and learning rate search space. This may suggest that self-attention and SSMs might operate quite similarly, which has been shown to be true under certain conditions [23].

While our work focuses on smaller-scale models, future work should increase the size of the models to confirm whether our findings persist. Analysis of metrics other than accuracy and certain training elements, such as optimizer, scheduler, and epochs, would also lead to better and more comprehensive results. Future work can explore applying a strong data augmentation regime to VIM-TI to see the impact on its performance and extending experiments to convolutional models. Overall, we hope this work can inspire more research to be conducted to determine why Mamba may perform better than Vision Transformers in imbalanced settings.

6. COMPLIANCE WITH ETHICAL STANDARDS

This is a numerical simulation study for which no ethical approval was required.

7. ACKNOWLEDGMENTS

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